Chapter 6 Answers

Problem 1

Note: Here is why the diagonal function is implied by the iteration theorem. Consider any $R_y(x, y, z)$. By the iteration theorem, there is some $\phi(y, z)$ such that $R_y(x, y, z)$ iff $R_{\phi(y,z)}(x)$. By definition of $R_i^n(x_1, ..., x_n)$ when $n = 1$, $R_{\phi(y,z)}(x)$ means $x \in \omega_{\phi(y,z)}$, and thus $x \in \omega_{\phi(y,z)}$ iff $R_y(x, y, z)$.

Consider any two recursively enumerable relations $M_1(x, y, z)$ and $M_2(x, y, z)$. Hence the relations $M_1(x, d(y, z), d(z, y))$ and $M_2(x, d(z, y), d(y, z))$ are recursively enumerable, and have indices m and n, respectively. Thus, if $d(y, z)$ is the diagonal function: $x \in \omega_{d(m,n)}$ iff $M_1(x, d(m, n), d(n, m))$ and $x \in \omega_{d(n,m)}$ iff $M_2(x, d(m, n), d(n, m))$, thus showing that $a = d(m, n)$ and $b = d(n, m)$ satisfies the theorem.

Problem 2

Consider any two recursively enumerable relations $M_1(x, y_1, y_2, z_1, z_2)$ and $M_2(x, y_1, y_2, z_1, z_2)$. Hence the relations $M_1(x, i, j, d(y, z), d(z, y))$ and $M_2(x, i, j, d(y, z), d(z, y))$ (as relations between x, y, z, i, j) are recursively enumerable for the diagonal function d. Hence, by the iteration theorem, there are recursive functions $t_1(i, j)$ and $t_2(i, j)$ such that for all numbers i, j, x, y, z:
(1): $R_{t_1(i,j)}(x, y, z)$ iff $M_1(x, i, j, d(y, z), d(z, y))$;
(2): $R_{t_2(i,j)}(x, y, z)$ iff $M_2(x, i, j, d(z, y), d(y, z))$.
Let $\phi_1(i, j)$ be the function $d(t_1(i, j), t_2(i, j))$ and $\phi_2(i, j)$ be the function $d(t_2(i, j), t_1(i, j))$. Thus $x \in \omega_{\phi_1(i,j)}$ iff $R_{t_1(i,j)}(x, t_1(i, j), t_2(i, j))$ iff $M_1(x, i, j, d(t_1(i, j), t_2(i, j)), d(t_2(i, j), t_1(i, j)))$ iff $M_1(x, i, j, \phi_1(i, j), \phi_2(i, j))$ and $x \in \omega_{\phi_2(i,j)}$ iff $R_{t_2(i,j)}(x, t_2(i, j), t_1(i, j))$ iff $M_2(x, i, j, d(t_1(i, j), t_2(i, j)), d(t_2(i, j), t_1(i, j)))$ iff $M_2(x, i, j, \phi_1(i, j), \phi_2(i, j))$. Thus $\omega_{\phi_1(i,j)} = \{x : M_1(x, i, j, \phi_1(i, j), \phi_2(i, j))\}$ and $\omega_{\phi_2(i,j)} = \{x : M_2(x, i, j, \phi_1(i, j), \phi_2(i, j))\}$, thus completing the proof.

Problem 3

(a): Let $R_1(x, y, y_1, z_1, z_2)$ be $M_1(x, y, z_1, z_2)$, and $R_2(x, y, y_1, z_1, z_2)$ be $M_2(x, y_1, z_1, z_2)$. By Theorem 2, there are recursive functions $\omega_{\phi_1(i,j)}$ and $\omega_{\phi_2(i,j)}$ such that:
(1) $\omega_{\phi_1(i,j)} = \{x : R_1(x, i, j, \phi_1(i, j), \phi_2(i, j))\} = \{x : M_1(x, i, \phi_1(i, j), \phi_2(i, j))\}$;
(2) $\omega_{\phi_2(i,j)} = \{x : R_2(x, i, j, \phi_1(i, j), \phi_2(i, j))\} = \{x : M_2(x, j, \phi_1(i, j), \phi_2(i, j))\}$
thus completing the proof of Theorem 3.

(b): Let $M(x, y, z_1, z_2)$ be the relation $R_y(x, z_1, z_2)$ (as a relation between x, y, z_1, z_2). Thus, by Theorem 3, there are recursive functions $\phi_1(i, j)$ and $\phi_2(i, j)$ such that:
(1) $\omega_{\phi_1(i,j)} = \{x : R_i(x, \phi_1(i, j), \phi_2(i, j))\}$;
(2) $\omega_{\phi_2(i,j)} = \{x : R_j(x, \phi_1(i, j), \phi_2(i, j))\}$
thus showing that Theorem 4 is a special case of Theorem 3.

Problem 4

Consider two recursively enumerable sets A and B. They are both Σ_1 and so there are Σ_0 relations $R_1(x, y)$ and $R_2(x, y)$ such that for all x:
(1) $x \in A$ iff $\exists y R_1(x, y)$;
(2) $x \in B$ iff $\exists y R_2(x, y)$.
Let the sets A' and B' be defined by the following two conditions:
(1) $x \in A'$ iff $\exists y(R_1(x, y) \wedge (\forall z \leq y) \sim R_2(x, z))$;
(2) $x \in B'$ iff $\exists y(R_2(x, y) \wedge (\forall z \leq y) \sim R_1(x, z))$.
Now suppose that $x \in A'$. Thus for some m:

$$(1)\ R_1(x, m) \wedge (\forall z \leq m) \sim R_2(x, z)$$

Suppose that $x \in B'$. Thus for some n:

$$(2)\ R_2(x, n) \wedge (\forall z \leq n) \sim R_1(x, z)$$

If $m \geq n$, then $\sim R_2(x, n)$ by (1), which is a contradiction. If $m < n$, then $\sim R_1(x, m)$ by (2), which is a contradiction. Thus if $x \in A'$, $x \notin B'$, and if $x \in B'$ then $x \notin A'$, thus proving that A' and B' are disjoint.

Problem 5

(1): Consider some disjoint pair of sets (A, B). We first assume that for any disjoint recursively enumerable supsets ω_i and ω_j of A and B respectively,

there is a number n outside both ω_i and ω_j. Now suppose that (A, B) was recursively separable under the first definition. Thus there is some recursive superset S of A disjoint from B. Thus both S and $\bar{S}$ are recursively enumerable, and $\bar{S}$ is a superset of B. Thus S and $\bar{S}$ are two recursively enumerable supersets of A and B respectively, but there are no numbers n outside both S and $\bar{S}$ (because they are complements), which is a contradiction. Thus (A, B) must be recursively inseparable under the first definition.

(2): Now suppose that (A, B) was recursively inseparable under the first definition. Hence there is no recursive superset of A disjoint from B. Now suppose that there were two disjoint recursively enumerable supersets ω_i and ω_j of A and B respectively, such that there was not a number n outside both ω_i and ω_j. Hence $\omega_i = \bar{\omega_j}$. Thus ω_i is a recursive superset of A disjoint from B, yielding a contradiction. Hence (A, B) must also be recursively inseparable under the second definition. Thus the proof that the two definitions are equivalent is done.

Problem 6

Suppose that $k(x, y)$ is a Kleene function for a disjoint pair (A_1, A_2). Thus:
(1) If $k(i, j) \in \omega_i - \omega_j$, then $k(i, j) \in A_2$;
(2) If $k(i, j) \in \omega_j - \omega_i$, then $k(i, j) \in A_1$.
Suppose that ω_i and ω_j are recursively enumerable supersets of A and B respectively. If $k(i, j) \in \omega_i$, then if $k(i, j) \notin \omega_j$, $k(i, j) \in A_2$ by (1). But this is impossible, as $A_2 \subseteq \omega_j$. Thus $k(i, j) \in \omega_j$. Now if $k(i, j) \in \omega_j$, then if $k(i, j) \notin \omega_i$, $k(i, j) \in A_1$ by (2). But this is impossible, as $A_1 \subseteq \omega_i$. Thus $k(i, j) \in \omega_i$. Hence $k(i, j) \in \omega_i$ iff $k(i, j) \in \omega_j$, showing that (A_1, A_2) is completely effectively inseparable under $k(x, y)$.

Problem 7

Let A_1 be the set of all numbers $\delta(i, j)$ such that $\delta(i, j) \in \omega_j$ (hence A_1 is recursively enumerable, in particular $x \in A_1$ iff $\exists i \exists j(\delta(i, j) = x \wedge x \in \omega_j)$). Let A_2 be the set of all $\delta(i, j)$ such that $\delta(i, j) \in \omega_i$ (hence A_2 is recursively enumerable, in particular $x \in A_2$ iff $\exists i \exists j(\delta(i, j) = x \wedge x \in \omega_i)$). Now suppose that $\delta(i, j) \in \omega_i - \omega_j$. Thus $x \in A_2 - A_1$. If $\delta(i, j) \in \omega_j - \omega_i$, then $x \in A_1 - A_2$. Thus $\delta(i, j)$ is a Kleene function for $(A_2 - A_1, A_1 - A_2)$. By the separation principle, we have two disjoint recursively enumerable sets K_1 and K_2 such that $A_2 - A_1 \subseteq K_1$ and $A_1 - A_2 \subseteq K_2$. Thus, if $\delta(i, j) \in \omega_i - \omega_j$, $x \in K_2$, and if $\delta(i, j) \in \omega_j - \omega_i$, then $\delta(i, j) \in K_1$. Hence $\delta(x, y)$ is a Kleene function for the

disjoint pair (K_1, K_2) of recursively enumerable sets.

Here is the more general result that if $\zeta(x, y)$ is a Kleene function for (B_1, B_2), and if B_1', B_2' are disjoint supersets of B_1, B_2 respectively, then $\zeta(x, y)$ must be a Kleene function for (B_1', B_2'). So suppose that $\zeta(x, y)$ is a Kleene function for (B_1, B_2), and that B_1', B_2' are disjoint supersets of B_1, B_2 respectively. Thus:
(1) If $\zeta(i, j) \in \omega_i - \omega_j$, then $\zeta(i, j) \in B_2$ and thus also in B_2';
(2) If $\zeta(i, j) \in \omega_j - \omega_i$, then $\zeta(i, j) \in B_1$ and thus also in B_1'.
Hence $\zeta(i, j)$ is a Kleene function for (B_1', B_2').

Problem 8

Suppose that $f(x)$ is the function which semi-reduces (A_1, A_2) to (B_1, B_2). Thus:
(1) If $x \in A_1$, then $f(x) \in B_1$;
(2) If $x \in A_2$, then $f(x) \in B_2$.
Let $k(x, y)$ be a Kleene function for (A_1, A_2). Thus:
(1') If $k(i, j) \in \omega_i - \omega_j$, then $k(i, j) \in A_2$;
(2') If $k(i, j) \in \omega_j - \omega_i$, then $k(i, j) \in A_1$.
By the iteration theorem, there is a recursive function $t(y)$ such that $x \in \omega_{t(y)}$ iff $f(x) \in \omega_y$. We now consider the function $f(k(t(i), t(j)))$. If $f(k(t(i), t(j))) \in \omega_i - \omega_j$, then $k(t(i), t(j)) \in \omega_{t(i)} - \omega_{t(j)}$, meaning, by (1'), $k(t(i), t(j)) \in A_2$. By (2), $f(k(t(i)), t(j)) \in B_2$. If $f(k(t(i), t(j))) \in \omega_j - \omega_i$, then $k(t(i), t(j)) \in \omega_{t(j)} - \omega_{t(i)}$, meaning, by (2'), $k(t(i), t(j)) \in A_1$. By (1), $f(k(t(i), t(j))) \in B_1$. Thus we have shown that the function $f(k(t(i), t(j)))$ is a Kleene function for (B_1, B_2).

Problem 9

Consider some disjoint semi-doubly-universal pair (U_1, U_2). We know that some Kleene pair exists, namely (K_1, K_2). Thus, by Proposition 2, because (K_1, K_2) is semi-reducible to (U_1, U_2) and U_1 and U_2 are disjoint, (U_1, U_2) is a Kleene pair. Then, by Proposition 1, (U_1, U_2) is completely effectively inseparable.

Problem 10

Suppose that (A, B) are doubly generative under some recursive function $\phi(x, y)$. Thus:
(1) $\phi(i, j) \in A$ iff $\phi(i, j) \in \omega_i$;
(2) $\phi(i, j) \in B$ iff $\phi(i, j) \in \omega_j$.
Consider some index e of the empty set. Now, we can see that the empty set has some index in many ways. For example, consider the sentence $P' \vdash P''$ having a Gödel number of p. Obvsioulsy, for every x, $px_0 \notin T_0$. Hence, for ω_p, because $x \in \omega_p$ iff $r(p, x) \in T_0$ iff $px_0 \in T_0$, ω_p is the empty set, as $px_0 \notin T_0$ for every x. Thus, because ω_i and ω_c are disjoint for every i, $\phi(i, c) \in A$ iff $\phi(i, c) \in \omega_i$. Hence A is generative under $\phi(i, c)$. In addition, B can be seen to be gerative under $\phi(c, i)$.

Problem 11

Suppose that the pair of recursively enumerable sets (A, B) is completely effectively inseparable under some recursive function $\zeta(x, y)$. By the iteration theorem, there are recursive function $t_1(i), t_2(i)$, such that for all x and i:
(1) $\omega_{t_1(i)} = \{x : x \in \omega_i \vee x \in A\}$;
(2) $\omega_{t_2(i)} = \{x : x \in \omega_i \vee x \in B\}$.
Let $\phi(i, j) = \zeta(t_1(j), t_2(i))$. Because (A, B) is completely effectively inseparable under $\zeta(x, y)$, if ω_i and ω_j are recursively enumerable supersets of A and B respectively, then $\zeta(i, j) \in \omega_i$ iff $\zeta(i, j) \in \omega_j$. Now suppose that ω_i and ω_j are disjoint. Thus $\zeta(t_1(j), t_2(i)) \in \omega_i$ iff $\zeta(t_1(j), t_2(i)) \in \omega_{t_2(i)}$ (by (2)) iff $\zeta(t_1(j), t_2(i)) \in \omega_{t_1(j)}$ iff $\zeta(t_1(j), t_2(i)) \in A$ (by (1) and the assumed disjointness of ω_i and ω_j). On the other hand, $\zeta(t_1(j), t_2(i)) \in \omega_j$ iff $\zeta(t_1(j), t_2(i)) \in \omega_{t_1(j)}$ (by (1)) iff $\zeta(t_1(j), t_2(i)) \in \omega_{t_2(i)}$ iff $\zeta(t_1(j), t_2(i)) \in B$ (by (2) and the assumed disjointness of ω_i and ω_j). Thus we have shown conditions (1) and (2) of being doubly generative are satisfied for (A, B) under the recursive function $\zeta(t_1(j), t_2(i))$.

Problem 12

Suppose that (A_1, A_2) is a disjoint pair doubly generative under the function $g(x, y)$. Thus we have conditions (1) and (2) of being doubly generative. For all numbers i and j:

C_1: Suppose that $\omega_i = \mathbb{N}$ and $\omega_j = \emptyset$. Thus $g(i, j) \in \omega_i$, and because

ω_i and ω_j are disjoint, $g(i,j) \in A_1$ by (1).

C_2: Suppose that $\omega_i = \emptyset$ and $\omega_j = \mathbb{N}$. Thus $g(i,j) \in \omega_j$, and because ω_i and ω_j are disjoint, $g(i,j) \in A_2$ by (2).

C_3: Suppose that $\omega_i = \omega_j = \emptyset$. Hence ω_i and ω_j are disjoint. Thus, if $g(i,j) \in A_1 \cup A_2$, $g(i,j) \in \omega_i \cup \omega_j$ by either (1) or (2), contrary to the asusmption that they are the empty set. Hence $g(i,j) \notin A_1 \cup A_2$.

Problem 13

Suppose that (A_1, A_2) is a disjoint pair and that $g(x,y)$ is a recursive functions such that C_1 and C_2 hold for (A_1, A_2). Now consider any disjoint pair (B_1, B_2) of recursively enumerable sets. By Problem 16 of Chapter 5, we have recursive functions $t_1(i), t_2(i)$ such that, for all i:
(1) If $i \in B_1$, then $\omega_{t_1(i)} = \mathbb{N}$;
(2) If $i \notin B_1$, then $\omega_{t_1(i)} = \emptyset$;
and
(1′) If $i \in B_2$, then $\omega_{t_2(i)} = \mathbb{N}$;
(2′) If $i \notin B_2$, then $\omega_{t_2(i)} = \emptyset$;
all hold. Let $f(x)$ be the recursive function $g(t_1(x), t_2(x))$.

(1): We first show that (B_1, B_2) is semi-reducible to (A_1, A_2) under $g(t_1(x), t_2(x))$. Suppose that $i \in B_1$. Thus $\omega_{t_1(i)} = \mathbb{N}$ by (1). If $\omega_{t_2(i)} \neq \emptyset$, then by (2′), $i \in B_2$, contrary to the assumption that B_1 and B_2 are disjoint. Thus $\omega_{t_2(i)} = \emptyset$, meaning $f(i) \in A_1$ by C_1. Similar reasoning can be used to prove that if $i \in B_2$, then $f(i) \in A_2$.

(2): Now assume that C_3 also holds for $g(x,y)$. We now show that if $i \notin B_1 \cup B_2$, then $i \notin A_1 \cup A_2$. So suppose that $i \notin B_1 \cup B_2$, meaning both $\omega_{t_1(i)} = \emptyset$ and $\omega_{t_2(i)} = \emptyset$ by (2) and (2′), respectively. Thus $g(t_1(i), t_2(i))$, that is $f(i)$, $\notin A_1 \cup A_2$ by C_3. Thus we have shown that the third condition of (B_1, B_2) being reducible to (A_1, A_2) holds. Because we showed the first two conditions held in part (1), we can conclude that (B_1, B_2) is reducible to (A_1, A_2) under $g(t_1(x), t_2(x))$.

It should be noted that the definition given for reducibility from one pair (B_1, B_2) to another pair (A_1, A_2) under some function $f(x)$ only means B_1 is reducible to A_1 and B_2 is reducible to A_2, both under $f(x)$, if A_1, A_2 are disjoint. For we are given the conditions:
(1) If $x \in B_1$, then $f(x) \in A_1$;

(2) If $x \in B_2$, then $f(x) \in A_2$;
(3) If $x \notin B_1 \cup B_2$, then $f(x) \notin A_1 \cup A_2$.
To show that B_1 is reducible to A_1 and B_2 is reducible to A_2, both under $f(x)$, we would need $x \in B_1$ iff $f(x) \in A_1$ and $x \in B_2$ iff $x \in A_2$. The first parts of both of those statements are given by (1) and (2) respectively. Suppose that $f(x) \in A_1$. By (3), it cannot be that $x \notin B_1 \cup B_2$. If $x \in B_2$, then $f(x) \in A_2$, contrary to the assumption that A_1 and A_2 are disjoint. Thus $x \in B_1$. Similar reasoning shows that if $f(x) \in A_2$, then $x \in B_2$. But this reasoning only works when A_1, A_2 are disjoint, so that assumption needs to be made.

Problem 14

Suppose that we have a doubly co-productive pair (A, B) under a recursive function $g(x, y)$. Thus, for all i and j, if ω_i is disjoint from ω_j, and ω_i is disjoint from A and ω_j disjoint from B, then $g(i, j)$ lies outside all the four sets ω_i, ω_j, A, B.

D_0: If $\omega_i = \omega_j = \emptyset$, then ω_i is disjoint from ω_j, and ω_i is disjoint from A and ω_j disjoint from B, meaing $g(i, j) \notin A \cup B$ by the asusmption of double co-productivity.

D_1: If $\omega_i = \{g(i, j)\}$ and $\omega_j = \emptyset$, then ω_i and ω_j are disjoint, and ω_j and B are disjoint. If $g(i, j) \notin A$, then ω_i and A would be disjoint, meaning, by the assumption of double co-productivity, $g(i, j) \notin \omega_i$, which is a contradiction. Thus $g(i, j) \in A$.

D_2: Similar reasoning can be used here as in the case of D_1.

Problem 15

Consider some recursive function $g(x, y)$. Let $M_1(x, y, z_1, z_2)$ be the recursively enumerable relation $x \in \omega_y \wedge x = g(z_1, z_2)$, and let $M_2(x, y, z_1, z_2)$ be the same relation. By Theorem 3, there are recursive functions $\phi_1(i, j)$ and $\phi_2(i, j)$ such that for all numbers i and j:
(1) $\omega_{\phi_1(i,j)} = \{x : M_1(x, i, \phi_1(i, j), \phi_2(i, j))\}$;
(2) $\omega_{\phi_2(i,j)} = \{x : M_1(x, j, \phi_1(i, j), \phi_2(i, j))\}$.
Take $t_1(i, j)$ and $t_2(i, j)$ to be $\phi_1(i, j)$ and $\phi_2(i, j)$, respectively. Thus $x \in \omega_{t_1(i,j)}$ iff $M_1(x, i, \phi_1(i, j), \phi_2(i, j))$ iff $x \in \omega_i \wedge x = g(\phi_1(i, j), \phi_2(i, j))$, which is the same as saying $\omega_{t_1(i,j)} = \omega_i \cap \{g(t_1(i, j), t_2(i, j))\}$. Also $x \in \omega_{t_2(i,j)}$ iff $M_2(x, j, \phi_1(i, j), \phi_2(i, j))$ iff $x \in \omega_j \wedge x = g(\phi_1(i, j), \phi_2(i, j))$, which is the same

as saying $\omega_{t_2(i,j)} = \omega_j \cap \{g(t_1(i,j), t_2(i,j))\}$.

Problem 16

Suppose that (A, B) is weakly doubly co-productive under $g(x, y)$. By the Lemma, there are recursive functions $t_1(x, y), t_2(x, y)$ such that for all i and j:
(1) $\omega_{t_1(i,j)} = \omega_i \cap \{g(t_1(i,j), t_2(i,j))\}$;
(2) $\omega_{t_2(i,j)} = \omega_j \cap \{g(t_1(i,j), t_2(i,j))\}$.
Let $\phi(x, y) = g(t_1(x, y), t_2(x, y))$. Because (A, B) is weakly doubly co-productive, $D_0 - D_2$ hold. Consider any two disjoint sets ω_i and ω_j. Because of their disjointness, $\omega_{t_1(i,j)}$ and $\omega_{t_2(i,j)}$ must also be disjoint.

We first show that if $\phi(i, j) \in \omega_i$, then $\phi(i, j) \in A$, and if $\phi(i, j) \in \omega_j$, then $\phi(i, j) \in B$. First, suppose that $\phi(i, j) \in \omega_i$. We know that $\phi(i, j) \in \omega_i$ iff $g(t_1(i, j), t_2(i, j)) \in \omega_i$ iff $g(t_1(i, j), t_2(i, j)) \in \omega_i \cap \{g(t_1(i, j), t_2(i, j)\} = \omega_{t_1(i,j)}$ by (1). Thus $\omega_{t_1(i,j)} = \{g(t_1(i, j), t_2(i, j)\}$. Now, if $\omega_{t_2(i,j)} \neq \emptyset$, then $\omega_{t_2(i,j)} = \{g(t_1(i, j), t_2(i, j)\}$ by (2), contrary to the assumption that $\omega_{t_1(i,j)}$ and $\omega_{t_2(i,j)}$ are disjoint. Thus $\omega_{t_2(i,j)} = \emptyset$, and therefore $g(t_1(x, y), t_2(x, y)) \in A$ by D_1. Similar reasoning can be used to conclude that if $\phi(i, j) \in \omega_j$, then $\phi(i, j) \in B$, replacing D_1 by D_2 and (2) and (1) by (1) and (2), respectively.

Next we show that if $\phi(i, j) \in A$, then $\phi(i, j) \in \omega_i$, and if $\phi(i, j) \in B$, then $\phi(i, j) \in \omega_j$. Now $\phi(i, j) \in A$ iff $g(t_1(i, j), t_2(i, j)) \in A$. By D_0, one of $\omega_{t_1(i,j)}$ or $\omega_{t_2(i,j)}$ must not be the empty set, or else $\phi(i, j) \notin A \cup B$, which would mean $\phi(i, j) \notin A$, a contradiction. So suppose that $\phi(i, j) \in \omega_{t_2(i,j)}$, which would imply that $\phi(i, j) \in \omega_j$ by (2). By the results in the previous paragraph, this would mean that $\phi(i, j) \in B$. But this is contrary to the assumption that A and B are disjoint. Thus $\phi(i, j) \in \omega_{t_1(i,j)}$, which implies that $\phi(i, j) \in \omega_i$ by (1). Similar reasoning can be used to show that if $\phi(i, j) \in B$, then $\phi(i, j) \in \omega_j$. Thus, we have shown that $\phi(i, j) \in \omega_i$ iff $\phi(i, j) \in A$ and $\phi(i, j) \in \omega_j$ iff $\phi(i, j) \in B$, meaning (A, B) is doubly generative under $\phi(i, j)$.

Problem 17

(a): Consider some recursively enumerable set A. Thus $x \in \omega_i \vee x \in A$ is also recursively enumerable. By the iteration theorem, there is some $t(i)$ such that $x \in \omega_{t(i)}$ iff $x \in \omega_i \vee x \in A$ iff $x \in \omega_i \cup A$.

(b): Now suppose that (A, B) is effectively inseparable under $\zeta(x, y)$

and that A and B are both recursively enumerable. By part (a), let $t_1(i), t_2(i)$ be recursive functions such that for all i, $\omega_{t_1(i)} = \omega_i \cup A$ and $\omega_{t_2(i)} = \omega_i \cup B$. Let $g(x, y) = \zeta(t_1(y), t_2(x))$. We consider any disjoint pair (ω_i, ω_j), and recall that A and B are disjoint. This means that $\omega_{t_1(j)}$ and $\omega_{t_2(i)}$ are disjoint, and thus, by the assumed effective inseparability of A and B, $g(i, j)$ lies outside both of $\omega_{t_1(j)}$ and $\omega_{t_2(i)}$, which implies that $g(i, j)$ lies outside all of ω_j, ω_i, A, B. This implies that:
(1) $g(i, j) \in \omega_i$ iff $g(i, j) \in A$;
(2) $g(i, j) \in \omega_j$ iff $g(i, j) \in B$.
Thus (A, B) is doubly generative under $g(x, y)$. In fact, (A, B) is also doubly generative under the function $g_1(x, y)$, where $g_1(x, y) = g(y, x)$.